

Cloud Storage Comparison

FleetCo Management LLC · Object storage options for dashcam media, uploads & fleet files · September 1, 2026

Executive summary

FleetCo currently stores **business records** in Render PostgreSQL (cloud) but saves **photos and dashcam JPEGs** on the web server disk (server/uploads/), which is not durable and does not scale. This document compares leading object storage providers to support a phased migration.

Recommendation: Move file/media storage to object storage immediately (AWS S3, Google Cloud Storage, or Cloudflare R2). Keep the app on Render + Postgres for now. Add 30–90 day dashcam retention. Upgrade database architecture only when entity count or performance requires it.

Current FleetCo storage (baseline)

Layer	Location	Durable?	Scales?
Users, customers, loads, vehicles, dashcam metadata	Render PostgreSQL (fleetco-db)	Yes	Moderate — single JSON blob in Postgres
JSON backup mirror	1 GB Render persistent disk (server/data)	Yes	Limited to 1 GB disk
Dashcam JPEGs, POD photos, receipts, PDFs	server/uploads/ (ephemeral web disk)	No	Poor — lost on redeploy, fills quickly
Driver offline queue	Phone IndexedDB / localStorage	Temporary	Until sync completes
Safety AI analysis	Gemini API (external) → alerts in Postgres	Yes (alerts only)	Depends on API limits

Dashcam math: At ~300 KB per JPEG and 1 frame/second, one driver generates ~1 GB/hour. Without retention policies, storage and cost grow rapidly on any platform.

Providers compared (2026 list pricing, US regions)

Factor	AWS S3	Google Cloud Storage	Cloudflare R2	Backblaze B2
Hot storage (\$/GB/mo)	~\$0.023	~\$0.020	~\$0.015	~\$0.007

Internet egress (\$/GB)	~\$0.09 (100 GB/mo free)	~\$0.12 (100 GB/mo free)	\$0.00	~\$0.01 (free up to 3× stored)
PUT requests (per 1K)	~\$0.005	~\$0.05	~\$4.50 per million Class A	Free
GET requests (per 1K)	~\$0.0004	~\$0.004	~\$0.36 per million Class B	Free
Archive / cold tier	Glacier Deep Archive ~\$0.001/GB	Archive ~\$0.0012/GB	Single tier only	Single hot tier
Lifecycle auto-delete	Yes — mature	Yes	Yes	Limited
Signed / private URLs	Yes	Yes	Yes	Yes
Market adoption / hiring	Industry standard	Strong (GCP / ML)	Growing (S3-compatible)	Niche / backup
S3-compatible API	Native	No (GCS API)	Yes	Yes

AWS S3 — benefits & trade-offs

Benefits

- Most widely used object store — largest ecosystem of tools, docs, and consultants
- Lower egress than GCS (~25% cheaper at scale) — important when fleet managers view dashcam in portal
- Eight storage classes (Standard → Glacier Deep Archive) for cost optimization
- Mature lifecycle rules — auto-delete dashcam after N days
- Natural path if FleetCo later adopts RDS, Lambda, or broader AWS stack
- Works from Render app via AWS SDK with minimal friction

Trade-offs

- IAM and bucket policies have a learning curve
- Total cost driven by egress + requests, not storage alone
- Separate vendor from Gemini (Google AI) — cross-cloud if AI reads from S3
- Does not fix Postgres JSON blob architecture — files only

Best for FleetCo when: you want the safest default, lower image delivery cost, and maximum portability.

Google Cloud Storage — benefits & trade-offs

Benefits

- Slightly cheaper hot storage (~\$0.020/GB) than S3 Standard
- Same Google Cloud console/billing as **Gemini Safety AI** (already in FleetCo stack)
- Easy pipeline: GCS image → Gemini vision without cross-cloud transfer fees
- Strong for future ML, BigQuery analytics on fleet data
- Clear tier ladder: Standard → Nearline → Coldline → Archive
- Enterprise-grade durability and IAM

Trade-offs

- Highest egress of the hyperscalers (~\$0.12/GB) — costly if portal loads many images
- Not S3-native — separate SDK vs AWS tooling
- Smaller third-party integration surface than S3
- Operation pricing can spike on list-heavy workloads

Best for FleetCo when: Google (Gemini, ML) is the long-term AI platform and egress volume stays moderate.

Cloudflare R2 — why it matters for FleetCo

R2 is often the best fit for **photo-heavy apps viewed often in a web portal** because it charges **\$0 egress** to the internet (via Cloudflare network). API is S3-compatible — minimal code change from an S3 integration.

Pros

Cons

Zero egress — fleet managers reviewing dashcam won't drive large bandwidth bills	Single storage tier — no Glacier-style deep archive
Lower storage than S3 (~\$0.015/GB)	Smaller ecosystem than AWS/GCP
S3-compatible — reuse AWS SDK patterns	Another vendor alongside Render + Postgres

Cost scenarios (estimated monthly, US list pricing)

Scenario	Stored	Egress (viewed)	AWS S3	GCS	Cloudflare R2
Early stage (light dashcam)	50 GB	20 GB/mo	~\$3	~\$3	~\$1
Growing fleet (moderate)	500 GB	200 GB/mo	~\$30	~\$34	~\$8
Heavy live dashcam (10 drivers, 2 hr/day each)	2 TB	800 GB/mo	~\$118	~\$136	~\$30
Heavy + no retention (worst case)	10 TB+	2 TB/mo	\$500+	\$600+	~\$150+

Estimates exclude API request fees and assume Standard/hot tiers. Retention policies dramatically reduce the "heavy" scenarios.

Decision matrix for FleetCo

Priority	Recommended choice	Reason
Fastest fix, lowest risk	AWS S3	Industry default, great docs, lifecycle rules, lower egress than GCS
Align with Gemini / Safety AI	Google Cloud Storage	Same cloud as vision AI; simpler future ML pipelines
Minimize portal viewing cost	Cloudflare R2	Zero egress — ideal for dashcam review UI
Cheapest raw storage	Backblaze B2	~\$0.007/GB but less enterprise polish
Keep everything on Render only	Not recommended	Uploads on ephemeral disk — data loss and crash risk
Full migration to AWS now	Defer	Overkill — move files first, app can stay on Render

Phased roadmap (recommended)

Phase	Timeline	Action	Outcome
1	Now	Object storage for all uploads (S3, GCS, or R2)	Durable media; survives redeploy
1	Now	Dashcam retention policy (30–90 days)	Predictable cost; compliance-friendly
2	3–6 mo	Upgrade Render Postgres + RAM as customers grow	Stable performance
3	6–12 mo	Split JSON blob or migrate to dedicated Postgres schema / RDS	Scales past thousands of entities
4	As needed	True video streaming (WebRTC) + queue-based AI	Only if live dashcam becomes core product

Security & compliance checklist

- **Private buckets** — no public read; serve via signed URLs with expiry
- **Encryption at rest** — enabled by default on S3 and GCS
- **Customer scoping** — prefix objects by customer_id / session
- **Retention** — document how long dashcam and cabin camera data is kept
- **Driver consent** — in-cabin camera requires clear fleet policy (varies by state)
- **Access control** — only authorized portal roles view Driver Media (already role-scoped)

One-line for leadership

"Our business data is already in cloud Postgres on Render. The gap is driver media — dashcam files on server disk won't scale and can be lost on redeploy. Move files to S3, GCS, or R2 with a 90-day retention policy now; keep the app on Render until we outgrow the database design."